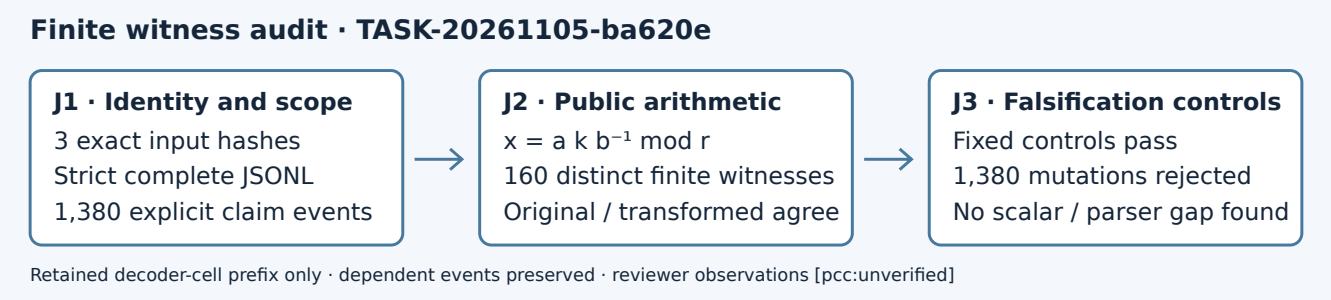
Finite public-witness arithmetic review

TASK-20261105-ba620e · 2026-11-05 · RUN-ECDLP-52de64

Reviewer observations [pcc:unverified]. All 1,380 scoped scalar claim events satisfy their stated original equations; all 460 RECOVERED word events also satisfy transformed equations and stored normalization. Their finite inventory contains 160 distinct (curve ID, target ID, scalar) entries. There are no invalid or undefined claimed witnesses, field disagreements, malformed lines, or unterminated tails in the supplied inputs. These are this reviewer's arithmetic observations, pending the Coordinator's native independence and publication checks.



Scope and evidence. The three complete input files match the frozen payload hashes and byte lengths: public input 72,966,009 bytes; traces 475,561,923 bytes / 36,782 LF records; decoder cells 864,461,632 bytes / 73,236 LF records. The explicit claim scope comprises 460 RECOVERED words, 460 nonnull cell scalars and 460 recovered_scalars entries. All 3,042,690 word records were traversed. Non-RECOVERED candidate words alone are outside the declared claim semantics. The 1,220 duplicate claim events remain linked to their cells, words, traces and blindings in checks.json. No recovery denominator or rate is inferred.

Derivation and implementation. From [k](aG) = bQ, multiplication by b inverse modulo prime r gives Q = [a k b inverse mod r]G. Therefore the stored original scalar must equal x = a k b inverse mod r. Semantics were written before candidate arithmetic. A fresh Python implementation uses extended Euclid, affine group addition and binary scalar multiplication. Canonical coordinates, nonzero G/Q, prime p/r, nonsingularity, rG = O, rQ = O and nonzero blinding residues were checked for every scoped witness domain. Exact trial division additionally confirms all scoped p/r (maximum p 7,691; maximum r 7,687). Producer arithmetic and public_checks flags were never used.

Controls and accounting. On $y^2 = x^3 + 2x + 2$ over F17, r = 19, G = (5,1), the positive 2G = (6,3) passes; 3G = (10,6) rejects that target, the wrong target (6,14) rejects 2G, and (0,0) is rejected as off-curve. Every original-valid claim's (x+1) mod r rejects the same Q (1,380 dependent mutation events). The audit performed 844 scalar multiplications and 4,489 equation checks, totaling 5,333; 3,620 multiplication cache hits are separately retained (8,953 request-inclusive upper bound). Peak measured Python RSS was 384,068 KiB, below 4 GiB. There were zero production runs.

Assigned joint verdicts. J1 holds: exact identity, parsing, joins and claim coverage. J2 holds: declared normalization, domains and independent equations. J3 holds: controls, mutations, event accounting and declared blind read. This makes no whole-algorithm or official scientific-status judgment. No producer code, sibling report, repository tree/history, solver, sampler, subgroup table or new randomness was read or run. Canonical graphs were outside the allowed input bundle and were not inspected or changed.

Primary sources, retrieved and independently read. SEC 1 v2.0 (2009), §2.2.1, https://www.secg.org/sec1-v2.pdf (SHA256 54687189…24a89); RFC 6090 (2011), §3 and Appendix F.1, https://www.rfc-editor.org/rfc/rfc6090.txt (SHA256 0066fc48…65a0). Exact hashes, source paths, public witnesses, parser details, full arithmetic command output, editable sources and accounting are in checks.json and verifier.py. The frozen external plan SHA256 is 23f44595bd0b6fb4f08b40125d8073d34ded9ea8e6a2ddb5e22edd49edda1a1e.